Central Ideas and Details — Part 2

Information and Ideas · Reading and Writing · 69 questions

2.1 8073f965-f7f2-4a4b-9a27-531088faf23a · Medium

Readers sometimes divide the works of twentieth-century English author Evelyn Waugh into two periods: one consisting of his early satirical novels and the other consisting of his later, more serious—even ponderous—books. Critic Seamus Perry, however, challenges that strict division. Perry argues that Waugh's writing didn't change over time as much as some readers have suggested. For instance, Perry contends that some of Waugh's earliest works, notably his biography of artist Dante Gabriel Rossetti, exhibit the earnest romanticism that would characterize Waugh's later fiction.

Based on the text, which statement about Waugh's works would Perry most likely agree with?

A. Waugh's works can appropriately be separated into two periods by their subject matter and tone.

B. Regardless of when they were written, Waugh's works have important similarities that transcend their differences.

C. The earliest of Waugh's works exhibit a satirical tone, even if that tone is more apparent in Waugh's later works.

D. Over time, Waugh's works became less humorous and more focused on weightier topics.

2.2 5676d961-5796-4b5b-93e0-d7b19a75f64a · Medium

The following text is adapted from Charles W. Chesnutt's 1901 novel *The Marrow of Tradition*.

> Mrs. Ochiltree was a woman of strong individuality, whose comments upon her acquaintance[s], present or absent, were marked by a frankness at times no less than startling. This characteristic caused her to be more or less avoided. Mrs. Ochiltree was aware of this sentiment on the part of her acquaintance[s], and rather exulted in it.

Based on the text, what is true about Mrs. Ochiltree's acquaintances?

A. They try to refrain from discussing topics that would upset Mrs. Ochiltree.

B. They are unable to spend as much time with Mrs. Ochiltree as she would like.

C. They are too preoccupied with their own concerns to speak with Mrs. Ochiltree.

D. They are likely offended by what Mrs. Ochiltree has said about them.

2.3 e830f97b-2162-4328-ad6d-23a86bafcd98 · Medium

In a study by Mika R. Moran, Daniel A. Rodriguez, and colleagues, residents of Quito, Ecuador, and Lima, Peru, were surveyed about parks in their cities. Of the 618 respondents from Quito, 82.9% indicated that they use the city's parks, and of the 663 respondents from Lima, 72.7% indicated using city parks. Given that the percentage of Quito respondents who reported living within a 10-minute walk of a park was much lower than that reported by Lima respondents, greater proximity alone can't explain the difference in park use.

The text makes which point about the difference between the proportions of Quito residents and Lima residents using parks?

A. It was much larger than the researchers conducting the study expected.

B. It is caused by something other than the parks' proximity to city residents.

C. It could be due to inaccuracies in the survey results.

D. It was calculated using sources that predate the survey.

2.4 70a3b3f6-b02a-4d56-93e8-aec50bc76eca · Medium

NASA's Aspera mission, led by Carlos Vargas, will investigate the circumgalactic medium (CGM), the huge swaths of low-density gas that fill and surround galaxies. Specifically, the team will focus on portions of the gas that exist in a "warm-hot" phase: these portions haven't previously been observable but are thought to fuel new star formation and hold most of the mass that makes up a galaxy. Using a telescope capable of revealing these parts of the CGM, the Aspera mission should help answer long-standing questions about how galaxies emerge, change, and even interact.

Which choice best states the main idea of the text?

A. As the leader of NASA's Aspera mission, Vargas will be the first person to investigate the makeup of the CGM.

B. Although galaxies that are surrounded by the CGM have been studied, researchers have been unable to directly observe low-density gas in the CGM in the "warm-hot" phase.

C. Researchers don't yet have a complete understanding of the process of galaxy evolution but have raised the possibility that galaxies interact with each other at times.

D. The Aspera mission is expected to produce the first direct observations of CGM gas in the "warm-hot" phase, which likely has an important role in the evolution of galaxies.

2.5 15643717-e244-4e2d-87ca-dc8aa43efc65 · Medium

Cats can judge unseen people's positions in space by the sound of their voices and thus react with surprise when the same person calls to them from two different locations in a short span of time. Saho Takagi and colleagues reached this conclusion by measuring cats' levels of surprise based on their ear and head movements while the cats heard recordings of their owners' voices from two speakers spaced far apart. Cats exhibited a low level of surprise when owners' voices were played twice from the same speaker, but they showed a high level of surprise when the voice was played once each from the two different speakers.

According to the text, how did the researchers determine the level of surprise displayed by the cats in the study?

A. They watched how each cat moved its ears and head.

B. They examined how each cat reacted to the voice of a stranger.

C. They studied how each cat physically interacted with its owner.

D. They tracked how each cat moved around the room.

2.6 2be296a8-a297-4cb0-9d75-002f50e1111e · Medium

The painter María Izquierdo played an important role in the development of twentieth-century Mexican art, but her work has never been well-known in the United States except among art historians. One reason for Izquierdo's relative obscurity is the enormous popularity of some of her peers. In particular, the painters Frida Kahlo and Diego Rivera have so captivated the interest of US audiences that Izquierdo and other Mexican artists from the period often get overlooked, despite the high quality of their work.

Which choice best states the main idea of the text?

A. Izquierdo's work is not as well-known in the United States as it should be because Kahlo and Rivera draw so much of the public's attention.

B. During Izquierdo's lifetime, her paintings were displayed in galleries in the United States much more frequently than paintings by Kahlo and Rivera were.

C. Izquierdo painted some of the same subjects that Kahlo and Rivera painted but used different techniques than they used.

D. Few of Izquierdo's works are in galleries today because she produced only a small number of paintings.

2.7 7073bd45-271c-4b31-b606-013a65b61485 · Medium

The recovery of a 1,000-year-old Chinese shipwreck in the Java Sea near present-day Indonesia has yielded a treasure trove of artifacts, including thousands of small ceramic bowls. Using a portable X-ray fluorescence analyzer tool, Lisa Niziolek and her team were able to detect the chemical composition of these bowls without damaging them. By comparing the chemical signatures of the bowls with those of the materials still at old Chinese kiln sites, Niziolek and her team can pinpoint which Chinese kilns likely produced the ceramic bowls.

Which choice best states the main idea of the text?

A. Because of a new technology, researchers can locate and recover more shipwrecks than they could in the past.

B. Researchers have been able to identify the location of a number of Chinese kilns in operation 1,000 years ago.

C. With the help of a special tool, researchers have determined the likely origin of bowls recovered from a shipwreck.

D. Before the invention of portable X-ray fluorescence, researchers needed to take a small piece out of an artifact to analyze its components.

2.8 57e0927e-845b-410d-a414-0ada4098e9b8 · Medium

The following text is adapted from Christina Rossetti's 1881 poem "Monna Innominata 2."

I wish I could remember that first day,

First hour, first moment of your meeting me,

If bright or dim the season, it might be

Summer or Winter for [all] I can say;

So unrecorded did it slip away,

So blind was I to see and to foresee,

So dull to mark the budding of my tree

That would not blossom yet for many a May.

Which choice best states the main idea of the text?

A. The speaker celebrates how the passage of time has strengthened a relationship that once seemed unimportant.

B. Because the speaker did not anticipate how important a relationship would become, she cannot recall how the relationship began, which she regrets.

C. As the anniversary of the beginning of an important relationship approaches, the speaker feels conflicted about how best to commemorate it.

D. After years of neglecting a once valuable relationship, the speaker worries it may be too late for her to salvage the relationship.

2.9 dcfca64d-f71d-4be1-8e85-40b6ef326b64 · Medium

In West Africa, jalis have traditionally been keepers of information about family histories and records of important events. They have often served as teachers and advisers, too. New technologies may have changed some aspects of the role today, but jalis continue to be valued for knowing and protecting their peoples' stories.

Which choice best states the main idea of the text?

A. Even though there have been some changes in their role, jalis continue to preserve their communities' histories.

B. Although jalis have many roles, many of them like teaching best.

C. Jalis have been entertaining the people within their communities for centuries.

D. Technology can now do some of the things jalis used to be responsible for.

2.10 c9b72f46-37cb-4778-ad91-d73d3d263723 · Medium

The following text is from Ezra Pound's 1909 poem "Hymn III," based on the work of Marcantonio Flaminio.

As a fragile and lovely flower unfolds its gleaming
foliage on the breast of the fostering earth, if
the dew and the rain draw it forth;
So doth my tender mind flourish, if it be fed with the
sweet dew of the fostering spirit,
Lacking this, it beginneth straightway to languish,
even as a floweret born upon dry earth, if the
dew and the rain tend it not.

Based on the text, in what way is the human mind like a flower?

A. It becomes increasingly vigorous with the passage of time.

B. It draws strength from changes in the weather.

C. It requires proper nourishment in order to thrive.

D. It perseveres despite challenging circumstances.

2.11 29cd0558-b798-46ac-8f8f-6fada55955cb · Medium

Culinary anthropologist Vertamae Smart-Grosvenor may be known for her decades of work in national public television and radio, but her book *Vibration Cooking: or, the Travel Notes of a Geechee Girl* is likely her most influential project. The 1970 book, whose title refers to Smart-Grosvenor's roots in the Low Country of South Carolina, was unusual for its time. It combined memoir, recipes, travel writing, and social commentary and challenged notions about conventions of food and cooking. Long admired by many, the book and its author have shaped contemporary approaches to writing about cuisine.

Which choice best describes the main idea of the text?

A. Smart-Grosvenor's unconventional book *Vibration Cooking: or, the Travel Notes of a Geechee Girl* is an important contribution to food writing.

B. Smart-Grosvenor held many different positions over her life, including reporter and food writer.

C. Smart-Grosvenor's groundbreaking book *Vibration Cooking: or, the Travel Notes of a Geechee Girl* didn't receive the praise it deserved when it was first published in 1970.

D. Smart-Grosvenor was a talented chef whose work inspired many people to start cooking for themselves.

2.12 0bc13525-918a-411b-bc8f-148fe9e12317 · Medium

Historians point to the rule of the Piast dynasty as crucial to the formation of the Polish state. However, some differentiate between members of the dynasty like Mieszko II Lambert, who ruled as king from 1025 to 1031 CE, and less well-documented figures like Siemomysł, who is said to have ruled in the 10th century but whose historical actuality is disputed. Siemomysł appears in the *Gesta principum Polonorum*, a chronicle of medieval Polish history written between 1112 and 1118. However, the chronicle's documentation of Siemomysł relies on oral tradition, unlike its records of later rulers.

According to the text, what is a difference between how historians view Siemomysł and how they view Mieszko II Lambert?

A. Historians agree that Mieszko II Lambert existed, but disagree about whether Siemomysł existed.

B. Historians believe that the *Gesta principum Polonorum* provides more evidence for Siemomysł's existence than it does for Mieszko II Lambert's existence.

C. Historians agree that Siemomysł ruled Poland much later than Mieszko II Lambert.

D. Historians find the orally transmitted stories affirming the existence of Mieszko II Lambert to be more convincing than similar stories about Siemomysł.

2.13 fa0a5216-8d0c-428d-8e98-a3f86ee9ba48 · Medium

Recently, scientists looked at data collected by NASA's InSight lander to learn more about seismic activity on Mars, known as marsquakes. The data show that the marsquakes all started from the same location on the planet. This discovery was surprising to scientists, as they expected that the marsquakes would originate from all over the planet because of the cooling of the planet's surface. Now, scientists believe that there could be areas of active magma flows deep beneath the planet's surface that trigger the marsquakes.

According to the text, what was surprising to scientists studying the seismic activity data from NASA's InSight lander?

A. The surface temperature of Mars has been rising.

B. There were different types of seismic waves causing marsquakes.

C. NASA's InSight lander collected less data than scientists had expected.

D. All the marsquakes started from the same location on the planet.

2.14 5e3760af-27dc-44b2-bf1c-53de6b88a3b6 · Medium

Algae living within the tissues of corals play a critical role in keeping corals, and the marine ecosystems they are part of, thriving. Some coral species appear brown in color when healthy due to the algae colonies living in their tissues. In the event of an environmental stressor, the algae can die or be expelled, causing the corals to appear white. To recover the algae, the bleached corals then begin to produce bright colors, which block intense sunlight, encouraging the light-sensitive algae to recolonize the corals.

What does the text most strongly suggest about corals that produce bright colors?

A. These corals have likely been subjected to stressful environmental conditions.

B. These corals are likely more vulnerable to exposure from intense sunlight than white corals are.

C. These corals have likely recovered from an environmental event without the assistance of algae colonies.

D. These corals are more likely to survive without algae colonies than brown corals are.

2.15 02287832-f99d-48ef-b1df-4b97c4d409b0 · Medium

The following text is from Anton Chekhov's 1898 short story "Ionitch" (translated by Marian Fell in 1915). The text is set in a Russian city referred to as the city of S.

> If newcomers to the little provincial city of S. complained that life there was monotonous and dull, its inhabitants would answer that, on the contrary, S. was a very amusing place, indeed, that it had a library and a club, that balls were given there, and finally, that very pleasant families lived there with whom one might become acquainted. And they always pointed to the Turkins as the most accomplished and most enlightened family of all.

What does the text suggest about the Turkins?

A. They are relative newcomers to the city of S.

B. They have a unique status in the city of S.

C. They have long disliked living in the city of S.

D. They are amused by the other residents of the city of S.

2.16 c2cf9861-143f-4813-a9dc-64c755159797 · Medium

In a study of new technology adoption, Davit Marikyan et al. examined negative disconfirmation (which occurs when experiences fall short of one's expectations) to determine whether it could lead to positive outcomes for users. The team focused on established users of "smart home" technology, which presents inherent utilization challenges but tends to attract users with high expectations, often leading to feelings of dissonance. The researchers found that many users employed cognitive mechanisms to mitigate those feelings, ultimately reversing their initial sense of disappointment.

Which choice best states the main idea of the text?

A. Research suggests that most users of smart home technology will not achieve a feeling of satisfaction given the utilization challenges of such technology.

B. Although most smart home technology is aimed at meeting or exceeding users' high expectations, those expectations in general remain poorly understood.

C. Research suggests that users with high expectations for a new technology can feel content with that technology even after experiencing negative disconfirmation.

D. Although negative disconfirmation has often been studied, little is known about the cognitive mechanisms shaping users' reactions to it in the context of new technology adoption.

2.17 fcc9d22f-ff02-4020-ae95-4a66720a6377 · Medium

The following text is adapted from Guy de Maupassant's nineteenth-century short story "The Trip of Le Horla" (translated by Albert M. C. McMaster, A. E. Henderson, Mme. Quesada, et al.). The narrator is part of a group traveling in a hot-air balloon at night.

> The earth no longer seems to exist, it is buried in milky vapors that resemble a sea. We are now alone in space with the moon, which looks like another balloon travelling opposite us; and our balloon, which shines in the air, appears like another, larger moon, a world wandering in the sky amid the stars, through infinity. We no longer speak, think nor live; we float along through space in delicious inertia. The air which is bearing us up has made of us all beings which resemble itself, silent, joyous, irresponsible beings, peculiarly alert, although motionless.

Which choice best states the main idea of the text?

A. The narrator feels a growing sense of isolation even though his companions are nearby during the balloon ride.

B. The narrator and his companions are completely absorbed in the change in perspective they gain while riding in the balloon.

C. The narrator and his companions are troubled by the disorienting effects of the altitude while riding in the balloon.

D. The narrator is pleasantly surprised by his companions' unrestrained enthusiasm about the sensation of riding in the balloon.

2.18 30511484-9757-45d7-958f-cf06ed690b94 · Medium

Believing that living in an impractical space can heighten awareness and even improve health, conceptual artists Madeline Gins and Shusaku Arakawa designed an apartment building in Japan to be more fanciful than functional. A kitchen counter is chest-high on one side and knee-high on the other; a ceiling has a door to nowhere. The effect is disorienting but invigorating: after four years there, filmmaker Nobu Yamaoka reported significant health benefits.

Which choice best states the main idea of the text?

A. Although inhabiting a home surrounded by fanciful features such as those designed by Gins and Arakawa can be rejuvenating, it is unsustainable.

B. Designing disorienting spaces like those in the Gins and Arakawa building is the most effective way to create a physically stimulating environment.

C. As a filmmaker, Yamaoka has long supported the designs of conceptual artists such as Gins and Arakawa.

D. Although impractical, the design of the apartment building by Gins and Arakawa may improve the well-being of the building's residents.

2.19 2f362274-9a69-425d-81d5-feb6db3a1e05 · Medium

To dye wool, Navajo (Diné) weaver Lillie Taylor uses plants and vegetables from Arizona, where she lives. For example, she achieved the deep reds and browns featured in her 2003 rug *In the Path of the Four Seasons* by using Arizona dock roots, drying and grinding them before mixing the powder with water to create a dye bath. To intensify the appearance of certain colors, Taylor also sometimes mixes in clay obtained from nearby soil.

Which choice best states the main idea of the text?

A. Reds and browns are not commonly featured in most of Taylor's rugs.

B. Taylor draws on local resources in the approach she uses to dye wool.

C. Taylor finds it difficult to locate Arizona dock root in the desert.

D. *In the Path of the Four Seasons* is widely acclaimed for its many colors and innovative weaving techniques.

2.20 6f567e67-97a4-4e24-8ae1-2bd68f0bdd7a · Medium

Online surveys are a common tool researchers use to collect information. These surveys are usually designed for use on personal computers (PCs), but more people are using smartphones to complete them than they are PCs. This shift in device usage may change how participants interact with online surveys. When researchers Jean Philippe Décieux and Philipp E. Sischka investigated, they found that PC users were more likely to multitask while taking surveys than smartphone users were, but PC users were also more likely to complete the surveys.

Which choice best states the main idea of the text?

A. People are choosing to take online surveys on smartphones more often than they are on PCs because smartphones are convenient.

B. Researchers are investigating why survey completion rates are higher on PCs than they are on smartphones, despite increased multitasking on PCs.

C. Researchers prefer online surveys to other ways of collecting information because they think online survey results are more reliable.

D. A study shows that the type of device people use to complete online surveys affects how they interact with these surveys.

2.21 f6d062f6-0ebb-4274-9345-3f4b6d44f52f · Medium

Microplastics are tiny pieces of plastic waste. Areas of the ocean with higher concentrations of microplastic particles also have smaller and fewer waves. A study by Yukun Sun and colleagues found that the concentration of microplastic particles cannot be the only reason for this reduced wave activity because the concentration of particles that would have the observed effect is much higher than that found in these areas of the ocean. However, they found that surfactants, chemicals often used to manufacture plastics, are released into the water from microplastics and have a much stronger wave-reducing effect.

According to the text, what did Sun and colleagues discover about surfactants?

A. They have a much stronger effect on wave activity than microplastics alone do.

B. They are mainly composed of water.

C. They are helpful for removing microplastics from the ocean.

D. They can be used to contain microplastics within certain areas of the ocean.

2.22 ed9e11c3-3d7d-468a-8256-17d36f4b3be6 · Medium

Given the immense scope of space, the search for extraterrestrial life is almost necessarily concentrated on the exoplanets deemed to have the most plausible chance of success—typically, atmosphere-bearing terrestrial planets orbiting within a certain range of their stars (termed the habitable zone). Claiming that Earth experienced a long transition from single-lid to plate tectonics that accelerated the emergence and evolution of complex organisms, researchers Robert J. Stern and Taras V. Gerya hold that consideration of tectonics, an often overlooked factor, could help further narrow the search for advanced extraterrestrial species.

Based on the text, what do Stern and Gerya most likely believe about the development of complex life on exoplanets?

A. It is more likely to occur on habitable zone planets with atmospheres and plate tectonics than on otherwise similar planets that lack plate tectonics.

B. It is more likely to occur if habitable zone planets with atmospheres transition from single-lid to plate tectonics late in their history than if they transition early in their history.

C. It is unlikely unless the transition from single-lid to plate tectonics occurs before the acquisition of a lasting atmosphere.

D. It is probably more dependent on the presence of plate tectonics than on orbital distance from a host star or the presence of an atmosphere.

2.23 dbf413aa-7ed0-4c04-a674-6de10190f3f5 · Medium

Many intellectual histories of the Black Power movement of the 1960s and 1970s rely heavily on essays and other explicitly ideological works as primary sources, a tendency that can overrepresent the perspectives of a small number of thinkers, most of whom were male. Historian Ashley D. Farmer has shown that expanding the array of primary sources to encompass more types of print material—including political cartoons, advertisements, and artwork—leads to a much better understanding of the movement and the crucial and diverse roles that Black women played in shaping it.

Which choice best describes the main idea of the text?

A. Farmer's methods and research have enriched the historical understanding of the Black Power movement and Black women's contributions to it.

B. Before Farmer's research, historians had largely ignored the intellectual dimensions of the Black Power movement.

C. Other historians of the Black Power movement have criticized Farmer's use of unconventional primary sources.

D. The figures in the Black Power movement whom historians tend to cite would have agreed with Farmer's conclusions about women's roles in the movement.

2.24 0597edea-9496-40db-983f-e9f02858badc · Medium

In her 1998 book *Blues Legacies and Black Feminism*, Angela Y. Davis bases her analysis in part on recordings of songs sung in the 1920s by Gertrude “Ma” Rainey and Bessie Smith. Davis focuses on how Rainey and Smith improvised the lyrics—replacing the original lines with mischievous jokes and wordplay. Davis’s work was particularly labor intensive because in order to transcribe, or write down, the lyrics as Rainey and Smith sang them, Davis had to listen repeatedly to the vinyl recordings, which weren’t very clear.

What does the text most strongly suggest about the songs sung by Rainey and Smith?

A. The songs have grown in popularity since Rainey and Smith first sang them.

B. There were more recordings made of Rainey’s songs than there were of Smith’s.

C. There were few, if any, reliable transcriptions of Smith’s and Rainey’s improvised lyrics when Davis began her research.

D. According to Davis, the songs sung by Rainey were more musically innovative than those sung by Smith typically were.

2.25 fdc309fb-203f-4eb0-ab14-ce679c48449c · Hard

The following text is adapted from María Cristina Mena's 1914 short story "The Vine-Leaf."

> It is a saying in the capital of Mexico that Dr. Malsufrido carries more family secrets under his hat than any archbishop.
>
> The doctor's hat is, appropriately enough, uncommonly capacious, rising very high, and sinking so low that it seems to be supported by his ears and eyebrows, and it has a furry look, as if it had been brushed the wrong way, which is perhaps what happens to it if it is ever brushed at all. When the doctor takes it off, the family secrets do not fly out like a flock of parrots, but remain nicely bottled up beneath a dome of old and highly polished ivory.

Based on the text, how do people in the capital of Mexico most likely regard Dr. Malsufrido?

A. Many have come to tolerate him despite his disheveled appearance.

B. Few feel concerned that he will divulge their confidences.

C. Some dislike how freely he discusses his own family.

D. Most would be unimpressed by him were it not for his professional expertise.

2.26 0d14ed05-7101-4b7c-9ad8-f817a8ee4705 · Hard

Many literary theorists distinguish between *fabula*, a narrative's content, and *syuzhet*, a narrative's arrangement and presentation of events. In the film *The Godfather Part II*, the *fabula* is the story of the Corleone family, and the *syuzhet* is the presentation of the story as it alternates between two timelines in 1901 and 1958. But literary theorist Mikhail Bakhtin maintained that *fabula* and *syuzhet* are insufficient to completely describe a narrative—he held that systematic categorizations of artistic phenomena discount the subtle way in which meaning is created by interactions between the artist, the work, and the audience.

Which choice best states the main idea of the text?

A. Literary theorist Mikhail Bakhtin argued that there are important characteristics of narratives that are not fully encompassed by two concepts that other theorists have used to analyze narratives.

B. Literary theorist Mikhail Bakhtin claimed that meaning is not inherent in a narrative but is created when an audience encounters a narrative so that narratives are interpreted differently by different people.

C. The storytelling methods used in *The Godfather Part II* may seem unusually complicated, but they can be easily understood when two concepts from literary theory are utilized.

D. Narratives that are told out of chronological order are more difficult for audiences to understand than are narratives presented chronologically.

2.27 79f2d4f0-39de-4a51-b567-a5d51d530f5c · Hard

For many years, the only existing fossil evidence of mixopterid eurypterids—an extinct family of large aquatic arthropods known as sea scorpions and related to modern arachnids and horseshoe crabs—came from four species living on the paleocontinent of Laurussia. In a discovery that expands our understanding of the geographical distribution of mixopterids, paleontologist Bo Wang and others have identified fossilized remains of a new mixopterid species, *Terropterus xiushanensis*, that lived over 400 million years ago on the paleocontinent of Gondwana.

According to the text, why was Wang and his team's discovery of the *Terropterus xiushanensis* fossil significant?

A. The fossil constitutes the first evidence found by scientists that mixopterids lived more than 400 million years ago.

B. The fossil helps establish that mixopterids are more closely related to modern arachnids and horseshoe crabs than previously thought.

C. The fossil helps establish a more accurate timeline of the evolution of mixopterids on the paleocontinents of Laurussia and Gondwana.

D. The fossil constitutes the first evidence found by scientists that mixopterids existed outside the paleocontinent of Laurussia.

2.28 0b043888-02d5-480f-bd62-7b4b70bef7a2 · Hard

Within higher education, studying philosophy requires that students be conversant with the field's foundational texts and historical figures. By contrast, *doing* philosophy within or beyond the academy demands the creative, self-directed application of acquired expertise to enduring questions about the nature of existence and knowledge. While both approaches engage with influential figures, those who do philosophy treat such figures as vital interlocutors who facilitate new insights rather than as ossified authorities who, though relevant to the present, primarily represent the discipline's past.

Based on the text, which choice best describes the relationship between doing philosophy and studying philosophy?

A. Doing philosophy helps students formulate concrete solutions to practical issues, whereas studying philosophy prioritizes engagement with historical arguments in the field.

B. Doing philosophy involves developing novel ideas through imagined dialogue with past philosophers based on knowledge of those philosophers' views acquired by studying philosophy.

C. Doing philosophy requires students to challenge the ideas articulated by past philosophers, especially when these ideas are broadly accepted by other people studying philosophy.

D. Doing philosophy represents a departure from the norms that govern scholarly inquiry, whereas studying philosophy requires conforming to these norms.

2.29 fd2f1d0f-fd4a-4ff6-892e-81e0dd9b2166 · Hard

Electronic music pioneer Wendy Carlos is credited with the music for three feature films: *A Clockwork Orange* (1971), *The Shining* (1980), and *Tron* (1982). However, her musical score for *A Clockwork Orange* is mostly made up of her arrangements of Ludwig van Beethoven's work. Also, almost all the music that she and Rachel Elkind composed for *The Shining* was unused by director Stanley Kubrick. It did not appear in the film. Of the three films, *Tron* is the one in which audiences can hear the most of Carlos's original compositions.

Which choice best states the main idea of the text?

A. Of the three films for which Carlos is credited, *Tron* features the most original music from her.

B. The director of *The Shining* used most of the music that Carlos composed for it.

C. Beethoven is widely considered to be a more important composer than Carlos.

D. Carlos is a notable innovator among film composers in the 1970s and 1980s.

2.30 3f72e0b2-8ad3-49cd-aa78-574c63fe17b8 · Hard

In a study of perceptions of listening ability, participants explained their position on a sociopolitical issue and then received a listener's written summary that expressed agreement or disagreement. Within selected summaries, researchers embedded markers of attentive listening (e.g., references to specific details), hypothesizing that such indications would positively influence perceptions of listening skill even in the context of disagreement. Instead, participants consistently rated listeners who expressed disagreement as less skilled, regardless of the other traits of the listeners' summaries.

What does the text most strongly imply about how participants responded to expressions of disagreement?

A. When participants felt personally invested in the topics they discussed, they were less likely to perceive listeners who expressed disagreement as attentive, regardless of evidence to the contrary in these listeners' summaries.

B. When participants encountered summaries from listeners who expressed disagreement with their views, participants tended to disregard evidence that the listeners had in fact been attentive.

C. Although participants maintained their positions regardless of a summary's level of detail, they tended to regard listeners who expressed disagreement as more attentive when these listeners provided more detailed summaries.

D. Although participants were critical of expressions of disagreement, they gave higher ratings to listeners whose summaries included markers of attentiveness than to listeners whose summaries did not include these markers.

2.31 fd02f4ff-6bf3-4885-92ba-dec5018038d6 · Hard

In a paper about p-i-n planar perovskite solar cells (one of several perovskite cell architectures designed to collect and store solar power), Lyndsey McMillon-Brown et al. describe a method for fabricating the cell's electronic transport layer (ETL) using a spray coating. Conventional ETL fabrication is accomplished using a solution of nanoparticles. The process can result in a loss of up to 80% of the solution, increasing the cost of manufacturing at scale—an issue that may be obviated by spray coating fabrication, which the researchers describe as "highly reproducible, concise, and practical."

What does the text most strongly suggest about conventional ETL fabrication?

A. It is less suitable for manufacturing large volumes of planar p-i-n perovskite solar cells than an alternative fabrication method may be.

B. It is more expensive when manufacturing at scale than are processes for fabricating ETLs used in other perovskite solar cell architectures.

C. It typically entails a greater loss of nanoparticle solution than do other established approaches for ETL fabrication.

D. It is somewhat imprecise and therefore limits the potential effectiveness of p-i-n planar perovskite solar cells at capturing and storing solar power.

2.32 376259b0-c855-4898-8e8e-9e8b363eca0f · Hard

The following text is from Maggie Pogue Johnson's 1910 poem "Poet of Our Race." In this poem, the speaker is addressing Paul Laurence Dunbar, a Black author.

Thou, with stroke of mighty pen,
  Hast told of joy and mirth,
And read the hearts and souls of men
  As cradled from their birth.

The language of the flowers,
  Thou hast read them all,
And e'en the little brook
  Responded to thy call.

Which choice best states the main purpose of the text?

A. To praise a certain writer for being especially perceptive regarding people and nature

B. To establish that a certain writer has read extensively about a variety of topics

C. To call attention to a certain writer's careful and elaborately detailed writing process

D. To recount fond memories of an afternoon spent in nature with a certain writer

2.33 90058464-79d7-422d-a91c-e84729445f02 · Hard

In their meta-analysis of research on advergames (video games developed to promote products or services), Zeph M.C. van Berlo et al. confirm that such games, though they can elicit player interest, may not facilitate subsequent recall of product and brand information. This phenomenon can be explained by the finite nature of cognitive capacity as it is articulated in Annie Lang's limited capacity model of motivated mediated message processing. In this case, players' cognitive resources are directed foremost toward the advergame's mechanics, leaving little or no capacity for encoding and storing the information the advertiser intends to be salient.

Which choice best states the main idea of the text?

A. The limited capacity model of motivated mediated message processing developed by Lang provides a means of explaining the finding by van Berlo et al. that players may not readily recollect the brand and product information embedded in advergames.

B. Research by van Berlo et al. corroborates Lang's conclusion that because people predominantly focus on game mechanics when playing video games, it is difficult for advergames to communicate brand and product information in ways that are highly memorable.

C. The meta-analysis by van Berlo et al. reveals that higher engagement in advergame mechanics is linked to lower effectiveness in persuading players to purchase particular brands and products.

D. Although the limited capacity model of motivated mediated message processing developed by Lang suggests otherwise, advergames can succeed as marketing tools, provided that they achieve a balance between game mechanics and the promotion of a brand or product.

2.34 fc1861ad-86ac-449d-9941-ad6dc31649dd · Hard

Optical tweezers are specialized scientific tools—particularly useful in biology and medicine—that use high-powered beams of light to trap and manipulate minuscule particles for study. Use of the tool has led to several scientific and medical breakthroughs over the last few decades, but the particles are often under prolonged exposure to the intense heat of the light beams. To overcome the risk of overheating, and thereby damage, researchers sometimes attach nano-sized glass beads to particles, allowing the light to focus on the beads instead of the particles.

Based on the text, what is one advantage of attaching glass beads to particles when using optical tweezers?

A. It decreases the time it takes for the optical tweezers to locate and capture the particles.

B. It facilitates the maneuvering of particles without directly heating the particles themselves.

C. It allows researchers to use weaker light beams to manipulate particles.

D. It adds a material to which particles can transfer any heat absorbed from the optical tweezers' light beam.

2.35 29cada86-74bb-41bb-a0dd-9a24a165a4e5 · Hard

The following text is from Milan Kundera's 1984 novel *The Unbearable Lightness of Being* (translated by Michael Henry Heim in 1984). Karenin is a dog that belongs to Tomas and Tereza.

> Karenin was not overjoyed by the move to Switzerland [from Prague]. Karenin hated change. Dog time cannot be plotted along a straight line; it does not move on and on, from one thing to the next. It moves in a circle like the hands of a clock, which—they, too, unwilling to dash madly ahead—turn round and round the face, day in and day out following the same path. In Prague, when Tomas and Tereza bought a new chair or moved a flower pot, Karenin would look on in displeasure. It disturbed his sense of time. It was as though they were trying to dupe the hands of the clock by changing the numbers on its face.

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Which choice best states the main idea of the text?

A. As a dog, Karenin possesses a sense of time that involves a strong preference for predictability and an aversion to disruption.

B. After he's moved to a new home, Karenin's negative response to changes has become more pronounced.

C. Similar to Tomas and Tereza, Karenin comprehends time as circular rather than as a straightforward progression.

D. As is the case for other dogs, Karenin's sense of time seems to accelerate depending on the objects and places that surround him.

2.36 12e22127-42e5-4d82-8b94-aa9cc152f77d · Hard

The following text is adapted from William Shakespeare's 1598 play *Henry IV, Part 1*. King Henry is addressing several noblemen who are rumored to have been disloyal to him.

> KING HENRY: My blood hath been too cold and temperate,
> Unapt to stir at these indignities,
> And you have found me, for accordingly
> You tread upon my patience; but be sure
> I will from henceforth rather be myself,
> Mighty and to be fear'd, than my [disposition],
> Which hath been smooth as oil, soft as young down,
> And therefore lost that title of respect
> Which the proud soul ne'er pays but to the proud.

Which statement best describes how King Henry presents himself in the text?

A. He declares that he has judged the noblemen's conduct in an impartial manner, and that despite their attempts to regain his trust through professions of submissiveness, they have irrevocably lost his respect.

B. He acknowledges to the noblemen that his tolerance of their conduct has undermined their respect for him, and he resolves to display his genuine nature, which is more forceful and compels deference.

C. He defends his calm disposition, which he sees as fundamental to his personality, and vows to maintain his tranquil demeanor even if it causes him to lose the respect of overly proud noblemen.

D. He concedes that he has treated the noblemen with indifference, which in part explains why they have defied him, but he asserts that their disrespectful behavior is primarily driven by their excessive pride.

2.37 2ed0236f-4451-4135-9977-0e08103aa3a2 · Hard

The following text is adapted from William Shakespeare's 1597 play *The Tragedy of King Richard III*. Richard is reflecting on the recent arrest of his brother, the Duke of Clarence, on suspicion of treason against King Edward IV. Derby, Hastings, Buckingham, Rivers, Dorset, and Grey are also members of the English nobility.

> RICHARD: I do the wrong, and first begin to brawl.
>
> The secret mischiefs that I set [flowing]
>
> I lay unto the grievous charge of others.
>
> Clarence, whom I indeed have cast in darkness,
>
> I do beweep to many simple [gullible people],
>
> Namely, to Derby, Hastings, Buckingham;
>
> And tell them 'tis the Queen and her allies
>
> That stir the King against the Duke my brother.
>
> Now they believe it, and withal whet me
>
> To be revenged on Rivers, Dorset, Grey.

Which choice best describes what happens in the text?

A. Richard describes having wept as he informed Derby, Hastings, and Buckingham that the queen and her allies convinced the king to act against Clarence, and says that the earnestness of his grief caused them to accept his version of events.

B. Richard attributes Clarence's troubles to both his own secret plotting and the distrust of Clarence that the queen and her allies Derby, Hastings, and Buckingham have planted in the king's mind.

C. Richard indicates that he has pretended to be aggrieved about Clarence's situation and has proclaimed it to be the fault of the queen and her allies, but in reality, he has caused the hostility the king feels toward Clarence.

D. Richard acknowledges that his mischievous nature has spurred him to commit misdeeds in the past, including instigating enmity between the king and Clarence, but he

reports that he has hitherto not lost the trust of the queen and her allies.

2.38 8544d09c-adf1-4069-8362-343c6a32fd8c · Hard

Some astronomers searching for extraterrestrial life have proposed that atmospheric NH_3 (ammonia) can serve as a biosignature gas—an indication that a planet harbors life. Jingcheng Huang, Sara Seager, and colleagues evaluated this possibility, finding that on rocky planets, atmospheric NH_3 likely couldn't reach detectably high levels in the absence of biological activity. But the team also found that on so-called mini-Neptunes—gas planets smaller than Neptune but with atmospheres similar to Neptune's—atmospheric pressure and temperature can be high enough to produce atmospheric NH_3.

Based on the text, Huang, Seager, and colleagues would most likely agree with which statement about atmospheric NH_3?

A. Its presence is more likely to indicate that a planet is a mini-Neptune than that the planet is a rocky planet that could support life.

B. Its absence from a planet that's not a mini-Neptune indicates that the planet probably doesn't have life.

C. It should be treated as a biosignature gas if detected in the atmosphere of a rocky planet but not if detected in the atmosphere of a mini-Neptune.

D. It doesn't reliably reach high enough concentrations in the atmospheres of rocky planets or mini-Neptunes to be treated as a biosignature gas.

2.39 938fef9c-6c98-458a-8ce2-236a09a80f22 · Hard

Paleontologists searching for signs of ancient life have found many fossilized specimens of prehistoric human ancestors, including several from the Pleistocene era discovered in a geological formation in the Minatogawa quarry in Japan. However, to study the emergence of the earliest multicellular organisms to appear on Earth, researchers must turn elsewhere, such as to the Ediacaran geological formation at Mistaken Point in Canada. A UNESCO World Heritage Site, the 146-hectare reserve contains more than 10,000 fossils that together document a critical moment in evolutionary history.

What does the text indicate about the geological formation at Mistaken Point?

A. It holds a greater number of fossils but from a smaller variety of species than the formation in the Minatogawa quarry does.

B. It has provided evidence that the earliest human species may have emerged before the Pleistocene era.

C. It is widely considered by paleontologists to be the most valuable source of information about prehistoric life forms.

D. It contains specimens from an older time period than those found in the formation in the Minatogawa quarry.

2.40 5d11fe17-72bb-49af-a3da-3be214ae0c9a · Hard

In 2018, scientists discovered an immense aggregation of *Muusoctopus robustus* (pearl octopuses) along a hydrothermal vent 3,200 meters beneath the ocean's surface. Water temperatures at this site—named the Octopus Garden—climb as high as 11°C, much warmer than the ambient 1.6°C typical at this depth. Based on observations made over three years, scientists concluded that temperatures at the site likely confer reproductive benefits and that the site is used exclusively for reproduction—6,000 *M. robustus* adults, hatchlings, and eggs were observed at the garden, but no juveniles were present.

Which statement about *M. robustus* and the Octopus Garden is best supported by the text?

A. *M. robustus* leave the Octopus Garden upon reaching an intermediary stage of development.

B. The *M. robustus* population at the Octopus Garden remains stable despite variations in water temperature.

C. *M. robustus* nests in the Octopus Garden contain on average fewer but larger eggs than nests at similar ocean depths.

D. The Octopus Garden provides an ideal feeding ground for *M. robustus* hatchlings.

2.41 44ff4a89-9f5d-4330-99f9-36c46b6585d5 · Hard

Conservationists worldwide are working to protect ecosystems from habitat destruction and biodiversity loss, and in many cases, initiatives that rely on natural features or processes can help address such challenges. In response to a rapidly dwindling population of blueback salmon, the Quinault Indian Nation (a tribe in Washington State) partnered with the conservation organization Wild Salmon Center to restore naturally occurring logjams in the Quinault River. The logjams create shady pools where the blueback salmon can rest and spawn, thus promoting blueback population recovery.

Which choice best states the main idea of the text?

A. A partnership between the Quinault Indian Nation and Wild Salmon Center shows the importance of collaborative approaches to preserving biodiversity.

B. Nature-based approaches can be effective ways to achieve conservation goals.

C. As indicated by a recent project, logjams help the blueback salmon thrive and reproduce.

D. Scientists now realize that nature-based conservation methods offer better long-term solutions to environmental issues than methods that are not nature-based do.

2.42 9f81443d-9195-4b30-907b-af16ce956e2e · Hard

Through adaptive radiation, multiple species of *Tetragnatha*—a genus of orb-weaving spiders—have significantly diversified their web architecture over a relatively short timescale. To investigate whether the material properties of *Tetragnatha* silk have been similarly affected by this evolutionary process, Angela M. Alicea-Serrano et al. examined the chemical composition of both the radial (non-sticky threads extending from the center) and capture (sticky threads that hold prey) silks that form the web as well as tested the tensile strength and viscosity of samples collected from three *Tetragnatha* species at two sites in the Hawaiian archipelago. The team found significant interspecies variation in these biomaterials.

Which choice best states the main idea of the text?

A. Research reveals that web architecture among species of *Tetragnatha* spiders has likely diversified more rapidly than the material properties of *Tetragnatha*'s silk have.

B. Research suggests that the material properties of silk from species of *Tetragnatha* spiders have diversified during adaptive radiation.

C. A study indicates that variations in web architecture among species of *Tetragnatha* spiders can likely be explained by corresponding variations in the material properties of their silks.

D. A study shows that adaptive radiation can explain interspecies variation in the web architecture of *Tetragnatha* spiders but not in the material properties of the spiders' silk.

2.43 7c58fd94-a98c-486c-aa23-2987cdca4de7 · Hard

Jürgen Kocka and other historians of capitalism rarely discuss domestic capitalism in Africa before the period of European colonization, implicitly presenting capitalism as external to and imposed on Africa. Crislayne Alfagali and other Africanist scholars have shown, however, that in parts of Africa, returns-focused investment, the establishment of open markets for wage labor, and other features of capitalism predated colonization. One reason for this discrepancy is that historians of capitalism tend to focus on longitudinal economic data drawn from archival records, which do not exist for much of precolonial Africa.

Which statement about Alfagali and other Africanist scholars is best supported by information in the text?

A. They likely make use of different types of evidence than historians of capitalism typically rely on.

B. They likely differ from historians of capitalism in the methods they use to derive longitudinal economic data from archival records.

C. They likely have a different view about which activities should be considered capitalist in nature than historians of capitalism do.

D. They likely view capitalism as having been more beneficial for Africa than historians of capitalism do.

2.44 b499ba18-43db-48ce-aa3c-8838c2104269 · Hard

In their 2022 paper, Christos Dimopoulos et al., having granted that the existence of antigravity—in which antimatter and matter repel rather than attract each other—lacked affirmative experimental support, rightly argued that such antigravity was worth considering on theoretical grounds given that evidence against it was similarly lacking. But a 2023 report by an international team of researchers details the first direct ballistic observations of antihydrogen atoms under gravity inside a CERN particle accelerator. Corresponding most closely to predictions under gravitational attraction, these observations were thoroughly inconsistent with antigravity.

Which choice best states the main idea of the text?

A. Antihydrogen ballistics observations were conducted at CERN to test specific conclusions about antigravity presented in the 2022 paper by Dimopoulos et al.

B. Although theoreticians were justified in studying antigravity before the release of the 2023 report, the report's findings suggest that the rationale for theoretical consideration offered in the 2022 paper by Dimopoulos et al. is no longer applicable.

C. The theoretical approach represented in the 2022 paper by Dimopoulos et al. assumed that unambiguous proof of antigravity would not be achievable, but the results in the 2023 report undermine that assumption.

D. Before 2023, researchers' inordinate focus on theoretical considerations hindered the development of the experimental regimen for direct antihydrogen ballistics observations.

2.45 6960936d-6e14-4293-8e40-181d7e56b513 · Hard

The following text is from Thomas Mann's 1924 novel *The Magic Mountain*, translated by John E. Woods in 1995.

> The story of Hans Castorp that we intend to tell here—not for his sake (for the reader will come to know him as a perfectly ordinary, if engaging young man), but for the sake of the story itself, which seems to us to be very much worth telling (although in Hans Castorp's favor it should be noted that it is *his* story, and that not every story happens to everybody)—is a story that took place long ago, and is, so to speak, covered with the patina of history and must necessarily be told with verbs whose tense is that of the deepest past.

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What does the text most strongly suggest about the story of Hans Castorp?

A. Though it is true that stories of even the most uninteresting people are themselves interesting because all people are unique, the reason this story is interesting is nonetheless difficult to understand because of the passage of time.

B. Even though it is a story of a person of no particular importance, its age and the manner in which it therefore must be told are both indicators that the story itself is important.

C. Like all stories about the lives of inconsequential people, this story must necessarily be related in a particular way if the reason the story is consequential is to be made evident to the audience.

D. It is a remarkable story that happened to an unremarkable person, though one could plausibly argue that because the story is valuable, some of its value accrues to the person at its center.

2.46 c62128d3-85ec-4b5a-8f7e-cedfc5850c29 · Hard

Elizabeth Asiedu has identified a negative correlation between the share of developing countries' economies derived from natural-resource extraction and those countries' receipts of foreign investment. This may appear counterintuitive—resource extraction requires initial investments (in extractive technology, for instance) at scales best met by multinational corporations—but Asiedu notes that natural-resource industries' boom-bust cycle can destabilize local currencies and increase developing countries' vulnerability to external shocks, creating levels of uncertainty to which foreign investors are typically averse.

Which choice best states the main idea of the text?

A. Although it may seem surprising that foreign investment declines in developing countries as natural-resource extraction makes up a larger share of those countries' economies, that decline happens because resource extraction requires initial investments too large for foreign investors to supply.

B. Although developing countries tend to become less dependent on foreign investment as natural-resource industries make up a larger share of their economies, this change may not occur if the boom-bust cycle of those industries destabilizes local currencies or increases countries' vulnerability to external shocks.

C. Although one might expect that foreign investment would increase as natural-resource extraction makes up a larger share of developing countries' economies, the opposite happens because heavy reliance on natural resources can lead to unattractive conditions for investors.

D. Although foreign investors tend to avoid initial investments in natural-resource industries in developing countries, foreign investment may increase significantly as those industries stabilize and the risks associated with them decline.

2.47 79be8588-88d3-4689-8dfa-a1eff7775048 · Hard

Studies of ocean wave breaking have predominantly focused on traveling waves (those propagating along the horizontal plane), so Mark McAllister et al. utilized a circular wave tank to produce and study spike waves, axisymmetric standing waves that can erupt vertically when traveling waves propagating in opposing directions intersect. Traveling waves break when wave steepness (height-to-length ratio) passes a critical threshold; breaking thus constrains wave height. McAllister et al. found that spike waves can exceed that constraint, as other factors than just steepness (e.g., jet stability and cavity shape) mediate spike-wave breaking.

Which choice best states the main idea of the text?

A. Previous studies have suggested that steepness mediates breaking in traveling waves, but the study by McAllister et al. shows that jet stability and cavity shape may also influence breaking in such waves.

B. The process of breaking limits the height of traveling waves, but the study by McAllister et al. suggests that spike waves can exceed those limits if their height-to-length ratio reaches a critical threshold.

C. McAllister et al. suggest that spike waves can form when traveling waves propagating in opposing directions intersect and that spike waves tend to be higher than traveling waves.

D. The study by McAllister et al. suggests that when traveling waves intersect in specific ways, the resulting wave may be higher than would be expected based on the properties of traveling waves.

2.48 0ab248c5-2e6c-4eb5-adb7-eec1beae9ae8 · Hard

To understand how temperature change affects microorganism-mediated cycling of soil nutrients in alpine ecosystems, Eva Kaštovská et al. collected plant-soil cores in the Tatra Mountains at elevations around 2,100 meters and transplanted them to elevations of 1,700–1,800 meters, where the mean air temperature was warmer by 2°C. Microorganism-mediated nutrient cycling was accelerated in the transplanted cores; crucially, microorganism community composition was unchanged, allowing Kaštovská et al. to attribute the acceleration to temperature-induced increases in microorganism activity.

It can most reasonably be inferred from the text that the finding about the microorganism community composition was important for which reason?

A. It provided preliminary evidence that microorganism-mediated nutrient cycling was accelerated in the transplanted cores.

B. It suggested that temperature-induced changes in microorganism activity may be occurring at increasingly high elevations.

C. It ruled out a potential alternative explanation for the acceleration in microorganism-mediated nutrient cycling.

D. It clarified that microorganism activity levels in the plant-soil cores varied depending on which microorganisms comprised the community.

2.49 e430357a-3c47-4449-815e-0422bc8f250a · Hard

Eighteenth-century economist Adam Smith is famed for his metaphor of the invisible hand, which he putatively used to illustrate a robust model of how individuals produce aggregate benefits by pursuing their own economic interests. Note “putatively”: as Gavin Kennedy has shown, Smith deploys this metaphor only once in his economic writings—to make a narrow point about the then-dominant economic theory of mercantilism—and it was largely ignored until some twentieth-century economists eager to secure an intellectual pedigree for their views elevated it to a fully-fledged paradigm.

Which choice best states the main idea of the text?

A. Although Smith is famed for his metaphor of the invisible hand, the metaphor was largely ignored until economists in the twentieth century came to realize that the metaphor was a robust model that anticipated their own views.

B. Some twentieth-century economists gave Smith’s metaphor of the invisible hand a significance it does not have in Smith’s work, but it is nevertheless a useful model of how individuals produce aggregate benefits by pursuing their own economic interests.

C. Smith’s metaphor of the invisible hand has been interpreted as a model of how individuals acting in their own interest produce aggregate benefits, but it was intended as a subtle critique of the economic theory of mercantilism.

D. The reputation of Smith’s metaphor of the invisible hand is not due to the importance of the metaphor in Smith’s work but rather to the promotion of the metaphor by some later economists for their own ends.

2.50 7eeec7e9-8081-478d-81de-6506deff4c1f · Hard

Hypothesizing that reliance on smartphones for information retrieval will lead a person to incorrectly remember instances of phone-assisted retrieval as instances of retrieval from memory, researchers asked participants general-knowledge trivia questions, instructing them to answer certain questions using only their smartphone and other questions using only their memory. When surveyed a week later, participants exhibited higher levels of misattribution for answers retrieved in the smartphone condition than for those retrieved in the memory condition, a finding that the researchers claimed supported their hypothesis.

Which question would be most useful to answer in determining the validity of the researchers' claim as it is presented in the text?

A. Were participants as likely to remember the source of their information when asked a week later as they would have been if asked immediately after answering the trivia questions?

B. Were participants less confident about the accuracy of information they were instructed to retrieve from a smartphone than they were of that retrieved from their own memory?

C. How likely were participants to answer all the trivia questions correctly in both the smartphone and the memory conditions?

D. How did participants attribute information they were instructed to retrieve from a smartphone in instances when they already knew that information independently of their phone?

2.51 4cdeef3b-c252-41b9-b816-e4e258e8d8e4 · Hard

Paleontologist Lucas E. Fiorelli and colleagues have reported the discovery at a mine in Brazil of several egg clutches, partially preserved single eggs, and egg shells from the Late Cretaceous period. The researchers have concluded that the area was once a nesting and breeding site for titanosaurs, a group of sauropod dinosaurs. The finding is significant given the previous lack of known nesting sites in northern regions of South America, which led many paleontologists to assume that titanosaurs migrated south to lay eggs.

What does the text most strongly suggest about the site discovered by the researchers?

A. It is the earliest known example of a titanosaur nesting and breeding site.

B. It was very difficult to excavate given that it was discovered in a mine.

C. It may have been occupied by other sauropods in addition to titanosaurs.

D. It is farther north than any other nesting site discovered in South America.

2.52 6beb6b1f-fb0d-4ffd-bf54-7e868f01ee48 · Hard

The following text is adapted from Lewis Carroll's 1889 satirical novel *Sylvie and Bruno*. A crowd has gathered outside a room belonging to the Warden, an official who reports to the Lord Chancellor.

> One man, who was more excited than the rest, flung his hat high into the air, and shouted (as well as I could make out) "Who roar for the Sub-Warden?" Everybody roared, but whether it was for the Sub-Warden, or not, did not clearly appear: some were shouting "Bread!" and some "Taxes!", but no one seemed to know what it was they really wanted.
>
> All this I saw from the open window of the Warden's breakfast-saloon, looking across the shoulder of the Lord Chancellor.
>
> "What can it all mean?" he kept repeating to himself. "I never heard such shouting before —and at this time of the morning, too! And with such unanimity!"

Based on the text, how does the Lord Chancellor respond to the crowd?

A. He asks about the meaning of the crowd's shouting, even though he claims to know what the crowd wants.

B. He indicates a desire to speak to the crowd, even though the crowd has asked to speak to the Sub-Warden.

C. He expresses sympathy for the crowd's demands, even though the crowd's shouting annoys him.

D. He describes the crowd as being united, even though the crowd clearly appears otherwise.

2.53 9881a760-f884-4181-a7f8-4ee2b2cb39a3 · Hard

In superfluorescence, electrical charges known as dipoles emit light in synchronized bursts so intense that they are visible to the eye. Until recently, this phenomenon has only been observed at extremely cold temperatures because dipoles cannot synchronize at higher temperatures. But in a study, Melike Biliroglu and colleagues observed superfluorescence at room temperature in thin films made of perovskite and other similarly crystalline materials; the researchers propose that the formation of shock-absorbing quasiparticles called polarons in the material protects dipoles from thermal interference.

Based on the text, how are polarons believed to be involved in the superfluorescence observed in Biliroglu and colleagues' study?

A. Polarons enable superfluorescent bursts to cross from one crystalline material to another.

B. Polarons allow for the dipoles to synchronize despite higher temperatures.

C. Polarons accelerate the dipoles' release of superfluorescent bursts.

D. Polarons decrease the intensity of the superfluorescent burst.

2.54 ed0c6123-ffe9-4c37-9b48-dfe7c470ff88 · Hard

Artificial leaves are a developing renewable energy technology that mimics the process of photosynthesis in plants. These devices are silicon-based solar cells coated in chemical catalysts that activate reactions that split water molecules into hydrogen and oxygen gas. The technology, while generating lots of interest, is not yet commercially viable as a large-scale energy source. To meet this challenge, scientists from many fields are researching ways to store, transport, and distribute the energy the devices produce while other scientists are working to improve the cost and efficiency of the devices.

Which choice best states the main idea of the text?

A. Continued research and development in artificial-leaf technology is needed before the devices can be widely used as an energy source.

B. The recent increase in the commercial use of artificial leaves as an energy source has encouraged many scientists to research ways to improve the technology.

C. Artificial leaves split water molecules into oxygen and hydrogen gas using catalysts more efficiently than plants do using the process of photosynthesis.

D. Artificial leaves were developed to mimic the natural process of photosynthesis in plants in order to store energy for long-term commercial use.

2.55 0c5e48fc-b8b5-4d2b-bef0-796bec56315c · Hard

The fynbos shrubland is a diverse habitat found only in South Africa. It is adjacent to the Afro-temperate forest, with almost no transition space between the two distinct habitats. Plants in the fynbos have uniquely thin and long root systems that spread out over large distances to absorb nutrients from the soil. Ecologists transplanted tree seedlings from the forest into plots in the fynbos. Seedlings in plots isolated from the roots of fynbos plants exhibited a growth rate five times greater than that of the seedlings in plots in close proximity to the roots of fynbos plants.

Based on the text, what role do fynbos roots most likely have in maintaining the border between the fynbos shrubland and the Afro-temperate forest habitats?

A. Fynbos roots damage the root systems of forest plants, leaving those plants unable to acquire sufficient nutrients.

B. Fynbos roots extend close enough to the forest plants' roots that they constitute a physical barrier that forest plants' roots cannot pass.

C. The root systems of fynbos plants allow the plants to take in so many soil nutrients that forest plants are prevented from flourishing in the fynbos.

D. The root systems of fynbos plants enhance the soil immediately surrounding the plants, allowing them to thrive in an otherwise harsh habitat.

2.56 b549c4fb-32ff-4538-9eee-d2a99127f351 · Hard

The following text is adapted from William Shakespeare's 1609 poem "Sonnet 27." The poem is addressed to a close friend as if he were physically present.

Weary with toil, I [hurry] to my bed,

The dear repose for limbs with travel tired;

But then begins a journey in my head

To work my mind, when body's work's expired:

For then my thoughts—from far where I abide—

[Begin] a zealous pilgrimage to thee,

And keep my drooping eyelids open wide,

What is the main idea of the text?

A. The speaker is asleep and dreaming about traveling to see the friend.

B. The speaker is planning an upcoming trip to the friend's house.

C. The speaker is too fatigued to continue a discussion with the friend.

D. The speaker is thinking about the friend instead of immediately falling asleep.

2.57 87a94542-6609-48ad-acd6-5ac706ccb0ba · Hard

Some animal-behavior studies involve observing wild animals in their natural habitat, and some involve capturing wild animals and observing them in a laboratory. Each approach has advantages over the other. In wild studies, researchers can more easily presume that the animals are behaving normally, and in lab studies, researchers can more easily control factors that might affect the results. But if, for example, the results from a wild study and a lab study of Western scrub-jays (*Aphelocoma californica*) contradict each other, one or both of the studies must have failed to account for some factor that was relevant to the birds' behavior.

Which choice best states the main idea of the text?

A. When the results of a natural-habitat study and those from a lab study of a wild animal such as the Western scrub-jay conflict, the study in the natural habitat is more likely than the lab study to have accurate results.

B. Studying wild animals such as the Western scrub-jay in both their natural habitat and lab settings is likely to yield conflicting results that researchers cannot fully resolve.

C. Wild animals such as the Western scrub-jay can be effectively studied in their natural habitat and in the lab, but each approach has drawbacks that could affect the accuracy of the findings.

D. Differing results between natural-habitat and lab studies of wild animals such as the Western scrub-jay are a strong indication that both of the studies had design flaws that affected the accuracy of their results.

2.58 55520f1a-8cea-4e4d-a5e4-12e755466da1 · Hard

The following text is from William Shakespeare's play *The Tempest*, first performed in 1611. Miranda has lived on an island with her father, Prospero, since she was three years old. Prospero has stated that Miranda likely does not remember anything other than her life on the island.

MIRANDA: 'Tis far off,

And rather like a dream than an assurance

That my remembrance warrants. Had I not

Four or five women once that tended me?

PROSPERO: Thou hadst, and more, Miranda. But how is it

That this lives in thy mind? What seest thou else

In the dark backward and abysm of time?

If thou remember'st ought ere thou camest here,

How thou camest here thou mayst.

In the text, which point does Prospero most directly make about Miranda and her memories?

A. Miranda's reminiscences about her early childhood have a melancholy quality that betrays her discontented view of her current circumstances.

B. Miranda's doubts about the accuracy of one recollection of a place other than the island are clouding her judgment and seem to be making her reluctant to explore her recollection of traveling to the island.

C. Miranda's ability to summon details of an experience she had before arriving on the island suggests that she may also be able to summon details of her arrival on the island.

D. Miranda's impression of a scene is vague because she is remembering a scenario she had daydreamed about as a child rather than a scenario that had occurred in reality.

2.59 de9b1463-1866-4898-8328-0c81e82e681b · Hard

Poetry in Classical Nahuatl, the language of the Aztec Empire, relies on *difrasismo*, or a parallel noun construction that conventionally operates as a single metaphor. For example, the common difrasismo *in cuauhtli in ocelotl* (literally, “the eagle, the jaguar”) signifies “warrior.” The device’s function is both formal—providing structure to lines of verse—and ritual: semantic relations among the two nouns and the concept they signify can be tenuous, as in the previous example, such that difrasismos are often only intelligible according to the conceptual associations observed in Aztec ceremonial culture.

Which statement about the difrasismo *in cuauhtli in ocelotl* is most strongly supported by the text?

A. Its metaphorical significance derives from the semantic equivalence of the two nouns constituting the difrasismo.

B. Its unintelligibility may cause its formal function within a line of verse to go unnoticed by present-day readers.

C. Its apparent obscurity can be resolved when considered in the proper cultural context.

D. Its frequency in Classical Nahuatl poetry confirms its intelligibility to the Aztec audience.

2.60 0fe52692-f446-41bb-85da-d4c4a0788f86 · Hard

The following text is Vita Sackville-West's circa 1920 poem "Evening." Spars are ships' masts, moorings are ropes that hold docked ships in place, and a riding-light is a light that a ship shines when it is anchored.

When little lights in little ports come out,

Quivering down through water with the stars,

And all the fishing fleet of slender spars

Range at their moorings, veer with tide about;

When race of wind is stilled and sails are furled,

And underneath our single riding-light

The curve of black-ribbed deck gleams palely white,

And slumbrous waters pool a slumbrous world;

—Then, and then only, have I thought how sweet

Old age might sink upon a windy youth,

Quiet beneath the riding-light of truth,

Weathered through storms, and gracious in retreat.

Which choice best states the main idea of the text?

A. The tranquility of a port in the evening can incline a person to appreciate the stillness of old age.

B. The difficulty of bringing a ship into port is apt training for dealing with the types of struggles encountered in old age.

C. A person who leads a long life that is varied and active may find it difficult to stay in a calm place.

D. The contrast between the peacefulness of a port at night and its activity during the day reflects the contrast between the calm of old age and the vibrancy of youth.

2.61 14f20e82-eb6c-41e2-a77c-78351a1cd67b · Hard

In episodes of high heat, corals expel symbiotic zooxanthellae from their tissues, leading to coral bleaching and heightened coral mortality. Some researchers have proposed a mitigation strategy of identifying, rearing, and disseminating corals that have evolved strong resistance to heat, but as Nia S. Walker et al. note, if genes conferring such resistance were exclusively advantageous, they should have already propagated through coral populations. Indeed, Walker et al. show that strongly heat-resistant corals display much less growth after heat stress than do moderately heat-resistant corals.

Based on the text, Walker et al. would most likely agree with which statement about the strategy some researchers have proposed?

A. It would involve promoting a trait that protects corals against high heat but that is associated with reduced thriving after episodes of high heat.

B. It would favor a trait that is advantageous now but would be disadvantageous if episodes of high heat become as infrequent as has been predicted.

C. It would benefit corals that are frequently exposed to episodes of high heat but would harm corals that are frequently exposed to episodes of moderate heat.

D. It would require manipulating a trait that is poorly understood and that may be linked to increased coral mortality during episodes of high heat.

2.62 ea65c400-ba2e-4387-9a23-fc9a727a5982 · Hard

Modern dog breeds are largely the result of 160 years of owners crossbreeding certain dogs in order to select for particular physical appearances. Owners often say that some breeds are also more likely than others to have particular personality traits—basset hounds are affectionate; boxers are easy to train—but Kathleen Morrill and colleagues found through a combination of owner surveys and DNA sequencing of 2,000 dogs that while physical traits are predictably heritable among purebred dogs, behavior varies widely among dogs of the same breed.

Which choice best states the main idea of the text?

A. Dog breeds would not exist without many years of human intervention in dogs' reproduction.

B. Research fails to confirm a commonly held belief about dog breeds and behavior.

C. The dog breeds most popular among owners have often changed over the past 160 years.

D. A study of dog breeds is notable for its usage of both opinion surveys and DNA sequencing.

2.63 ea859268-8f45-40e6-add4-ecd3844151f4 · Hard

Disco remains one of the most ridiculed popular music genres of the late twentieth century. But as scholars have argued, the genre is far less superficial than many people believe. Take the case of disco icon Donna Summer: she may have been associated with popular songs about love and heartbreak (subjects hardly unique to disco, by the way), but like many Black women singers before her, much of her music also reflects concerns about community and identity. These concerns are present in many of the genre's greatest songs, and they generally don't require much digging to reveal.

What does the text most strongly suggest about the disco genre?

A. It has been unjustly ignored by most scholars despite the importance of the themes addressed by many of the genre's songs.

B. It evolved over time from a superficial genre focused on romance to a genre focused on more serious concerns.

C. It has been unfairly dismissed for the inclusion of subject matter that is also found in other musical genres.

D. It gave rise to a Black women's musical tradition that has endured even though the genre itself faded in the late twentieth century.

2.64 50ef9940-0c0d-4d71-a837-c302dfa17130 · Hard

The following text is adapted from Fanny Burney's 1778 novel *Evelina*. The text is an excerpt from a letter by Lady Howard to Evelina's guardian, Reverend Villars, in which Lady Howard discusses an upcoming trip to London planned by her daughter Mrs. Mirvan's family.

> It is very earnestly [the Mirvans'] wish to enlarge and enliven their party by the addition of your amiable ward, who would share, equally with her own daughter, the care and attention of Mrs. Mirvan. Do not start at this proposal; it is time that [Evelina] should see something of the world. When young people are too rigidly sequestered from it, their lively and romantic imaginations paint it to them as a paradise of which they have been beguiled; but when they are shown it properly, and in due time, they see it such as it really is, equally shared by pain and pleasure, hope and disappointment.

Based on the text, Lady Howard would most likely agree with which statement about Reverend Villars?

A. Although the manner in which he has raised Evelina is in many ways exemplary, he has been misguided in shielding her from the influence of other young people.

B. He has imparted to Evelina his own idealistic view of the world, which results in her being unprepared to face inevitable disappointments.

C. Although his desire to guard Evelina from unscrupulous people is commendable, his general mistrust has led him to be unduly wary of the Mirvans.

D. He is overly protective of Evelina, who would likely benefit from a greater variety of experiences than she has had thus far.

2.65 b7b895db-15c3-4714-912a-fbc70f8d95ce · Hard

In 2019, 20 previously unknown moons were confirmed to be orbiting Saturn. Three of the moons have prograde orbits (orbiting in the direction the planet spins), and the other 17 have retrograde orbits (orbiting in the opposite direction of the planet's spin). All but one of the 20 moons are thought to be remnants of bodies that orbited Saturn until they broke apart in collisions. Although the one exceptional moon orbits in the same direction as the planet's spin, its orbit is highly eccentric compared to the rest, which may suggest that it has a different origin than the other 19 moons.

Based on the text, which choice best describes the moon with the eccentric orbit?

A. It doesn't have a retrograde orbit, but it likely has the same origin as the moons with retrograde orbits.

B. Its orbit is so tilted with respect to the other moons' orbits that it's neither prograde nor retrograde.

C. It has a prograde orbit that is likely the result of having collided with another body orbiting Saturn.

D. It has a prograde orbit and may not be a remnant of an earlier body that orbited Saturn.

2.66 980ff1d5-d247-4ec5-a99c-63476cc5503b · Hard

The following text is from Mick Herron's 2023 novel *The Secret Hours*. The narrator is describing members of a government committee.

> Finally, and adding much-needed gravitas, was Sir Winston Day, whose features seemed moulded to adorn a bust, or possibly a stamp, and whose forehead was so evidently bulging with grey matter that it would have been impertinent to inquire too closely into the actual achievements his half century of public service had produced. His recently published memoirs possibly cast light on this enigma, but given that such details were not provided until after the thirty-page mark, they might as well have remained state secrets.

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Based on the text, which choice best describes Sir Winston Day?

A. He has the appearance of a distinguished figure, but it is uncertain whether he has accomplished anything to earn distinction.

B. He looks like a person worthy of respect, but his memoirs reveal that some of his actions were dishonorable.

C. He would be a celebrated public figure if his achievements did not have to be kept secret.

D. He has maintained a modest profile even though he has served the public capably for many years.

2.67 58ef0410-5330-4262-9361-987ad9d1689e · Hard

Philosophers note that many people have an intuitive sense that while we ought not to lie, there may be circumstances in which lying is permissible. If this intuition is correct and we lack an inviolable duty to speak truthfully, what grounds opposition to lying in the first place? Japa Pallikkathayil has advanced one answer by appealing to a duty to respect others' agential interests: the possession of false beliefs constrains agency, and thus we ought not to impede the formation of true beliefs unless doing so prevents a greater constraint on someone's agency or an otherwise impermissible end.

Which choice best states the main idea of the text?

A. One potential means of justifying opposition to lying is Pallikkathayil's argument that we have an obligation to respect other people's agency that entails a commitment to truthfulness except in certain circumstances.

B. Many people have an intuitive sense that lying is permissible in some circumstances but lack a principled way to identify those circumstances, and Pallikkathayil's argument may provide a means of resolving that problem.

C. Pallikkathayil's argument suggests that if we have a duty to respect other people's agential interests and if possession of false beliefs constrains agency, then we have an inviolable duty to speak truthfully.

D. Pallikkathayil's argument shows that if our intuition that circumstances may make lying permissible is correct, then it is unclear whether there are any grounds for an opposition to lying in the first place.

2.68 9143317f-29a7-4559-930f-3895f4457c7d · Hard

The most recent iteration of the immersive theater experience *Sleep No More*, which premiered in New York City in 2011, transforms its performance space—a five-story warehouse—into a 1930s-era hotel. Audience members, who wander through the labyrinthine venue at their own pace and follow the actors as they play out simultaneous, interweaving narrative loops, confront the impossibility of experiencing the production in its entirety. The play's refusal of narrative coherence thus hinges on the sense of spatial fragmentation that the venue's immense and intricate layout generates.

What does the text most strongly suggest about *Sleep No More*'s use of its performance space?

A. The choice of a New York City venue likely enabled the play's creators to experiment with the use of theatrical space in a way that venues from earlier productions could not.

B. Audience members likely find the experience of the play disappointing because they generally cannot make their way through the entire venue.

C. The production's dependence on a particular performance environment would likely make it difficult to reproduce exactly in a different theatrical space.

D. Audience members who navigate the space according to a recommended itinerary will likely have a better grasp of the play's narrative than audience members who depart from that itinerary.

2.69 3bfe8f14-8d16-47a6-ac4c-e7ddb1572611 · Hard

Several scholars have argued that conditions in England in the late ninth through early eleventh centuries—namely, burgeoning literacy amid running conflicts between England's Anglo-Saxon kingdoms and Danish invaders—were especially conducive to the production of the Old English epic poem *Beowulf*, and they have dated the poem's composition accordingly. It is not inconceivable that *Beowulf* emerged from such a context, but privileging contextual fit over the linguistic evidence of an eighth- or even seventh-century composition requires a level of justification that thus far has not been presented.

Which choice best states the main idea of the text?

A. Although there are some grounds for believing that *Beowulf* was composed between the late ninth and early eleventh centuries, advocates for that view tend to rely on evidence that has been called into question by advocates for an earlier date.

B. Although several scholars have dated *Beowulf* to the late ninth through early eleventh centuries, others have argued that doing so privileges a controversial interpretation of the social conditions of the period.

C. Although *Beowulf* fits well with the historical context of England in the late ninth through early eleventh centuries, it fits equally well with the historical context of England in the seventh and eighth centuries.

D. Although the claim of a late ninth- through early eleventh-century composition date for *Beowulf* has some plausibility, advocates for the claim have not compellingly addressed evidence suggesting an earlier date.